

JEE-Main-21-01-2026 (Memory Based)

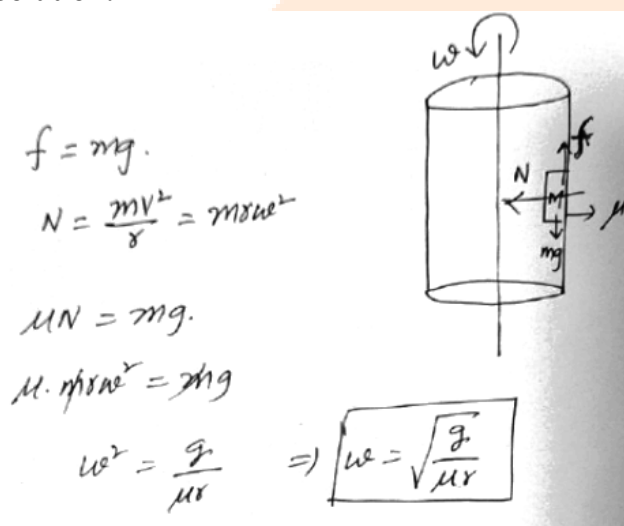
[EVENING SHIFT]

Physics

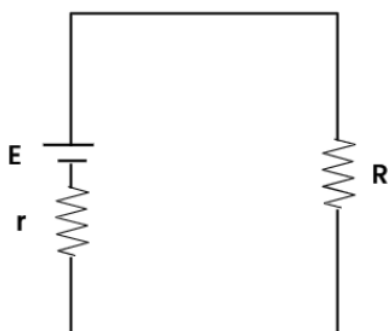
Question: A big drum of radius 'r' had a block of mass 'm' stuck to its wall & it has coeff of friction ' μ ' & rotating with ' ω ' rad/s. For what value of ' ω ' will the block stick to the wall?

Answer: ($\omega = \sqrt{\frac{g}{\mu r}}$)

Solution:



Question: In a circuit there is a battery with internal resistance r and Emf E , which is connected to external load resistance R as shown. Find value of R so that maximum power dissipated across R .

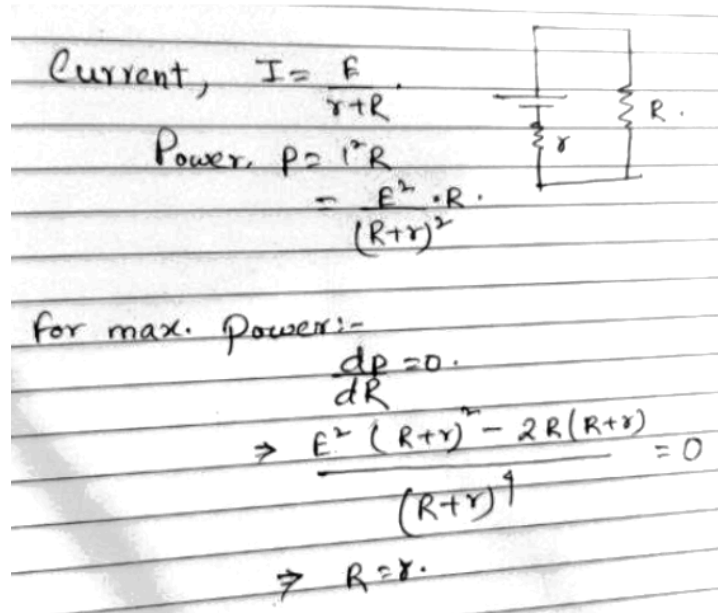


Options:

- (a) $R = r$
- (b) $R = r/2$
- (c) $R = \sqrt{2} r$
- (d) $R = 2r$

Answer: (a)

Solution:



Current, $I = \frac{E}{r+R}$

Power, $P = I^2 R = \frac{E^2 \cdot R}{(R+r)^2}$

for max. power:-

$$\frac{dP}{dR} = 0$$

$$\Rightarrow \frac{E^2 (R+r)^{-2} - 2R(R+r)^{-3}}{(R+r)^4} = 0$$

$$\Rightarrow R = r$$

Question: A spherical ball of radius 6 mm falls through a viscous fluid and attains terminal velocity of 20 m/s. If another spherical ball of same material and density but of radius 3 mm falls through the same fluid, Find V_T .

Answer: (5 m/s)

Solution:

$$V_T \propto r^2$$

$$r_1 = 6 \text{ mm}, V_1 = 20 \text{ m/s}$$

$$r_2 = 3 \text{ mm}, V_2 = ?$$

$$\frac{V_2}{V_1} = \left(\frac{r_2}{r_1}\right)^2 = \left(\frac{3}{6}\right)^2 = \frac{1}{4}$$

$$\Rightarrow V_2 = V_1 \times \frac{1}{4} = 20 \times \frac{1}{4} = 5 \text{ m/s}$$

Question: A particle of mass $m = 9.1 \times 10^{-31} \text{ kg}$ and charge $e = 1.6 \times 10^{-19} \text{ C}$ is accelerated through a potential difference of 1.2 V. If $h = 6.63 \times 10^{-34} \text{ J s}$, Find its de-Broglie wavelength.

Answer: (11.21 nm)

Solution:

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2meV}}$$

$$= \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} \times 1.2}}$$

Question: What will be significant figure of summation of 0.153, 153.2 and 1532?

Options:

- (a) 3
- (b) 4
- (c) 5
- (d) 6

Answer: (b)

Solution:

Handwritten calculation:

$$\begin{array}{r} 0.153 \\ 153.2 \\ 1532 \\ \hline 1685.353 \end{array}$$

∴ limiting term has 0 decimal J.

Significant fig: - 4.

Question: The energy of an electron in nth bohr orbit is given by $E_n = -\frac{\epsilon_0}{n^2}$. For a certain excited state of hydrogen, the magnitude of electron's energy is formed to be

$\frac{\epsilon_0}{25}$. Find value of $\frac{2\pi L}{h}$.

Answer: (5)

Solution:

$$E_n = \frac{\varepsilon_0}{n^2}$$

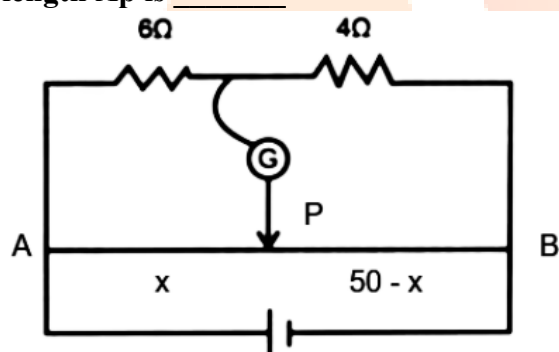
$$\Rightarrow 0.04 \varepsilon_0 = \frac{\varepsilon_0}{n^2}$$

$$\Rightarrow \boxed{n=5}$$

$$\therefore L = \frac{n h}{2\pi}$$

$$\boxed{n = \frac{2\pi L}{h} = 5}$$

Question: The total length of potentiometer wire AB is 50 cm in the arrangement shown in the in fig. If 'P' is the point where the galvanometer shows zero reading Then the length Ap is _____



Options:

- (a) 15
- (b) 20
- (c) 30
- (d) 25

Answer: (c)

Solution:

Sol:-

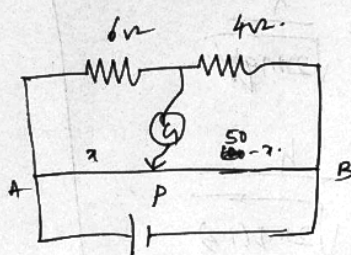
From wheatstone bridge concept.

$$\frac{6}{2} = \frac{4}{50-x}$$

$$300 - 6x = 4x$$

$$10x = 300$$

$$x = 30 \Omega$$



Question: The Kinetic energy of a simple harmonic oscillating with angular frequency of 176 rad/s. The frequency of the simple harmonic oscillator is -

$$Hz \left(\pi = \frac{22}{7} \right)$$

Options:

- (a) 176
- (b) 14
- (c) 28
- (d) 88

Answer: (d)

Solution:

$$K.E = \frac{1}{2} m \omega^2 A^2 \cos^2 \omega t$$

$$= \frac{1}{4} \times m \omega^2 A^2 (1 + 2 \cos \omega t)$$

$$K.E \rightarrow \omega = 176 \text{ rad/s}$$

$$SHM \rightarrow \frac{\omega}{2} = 88 \text{ rad/s}$$

Question: Two spheres having equal mass m , charge q and radius R , are moving towards each other. Both have speed u at an instant when distance between their centers is $4R$. Minimum value of u so that they touch each other is

Options:

(a) $\sqrt{\frac{q^2}{4\pi\epsilon_0 m R}}$

(b) $\sqrt{\frac{q^2}{16\pi\epsilon_0 m R}}$

(c) $\sqrt{\frac{q^2}{\pi\epsilon_0 m R}}$

(d) $\sqrt{\frac{q^2}{8\pi\epsilon_0 m R}}$

Answer: (b)

Solution:

$\Rightarrow \frac{1}{2}mu^2 + \frac{1}{2}mu^2 - 0 = \frac{kq^2}{2R} - \frac{kq^2}{4R}$
 $\Rightarrow mu^2 = \frac{kq^2}{4R}$
 $\Rightarrow u^2 = \frac{q^2}{16\pi\epsilon_0 m R}$
 $\therefore u = \sqrt{\frac{q^2}{16\pi\epsilon_0 m R}}$

Question: The rms speed of O_2 molecules at $47^\circ C$ is equal to rms speed of H_2 molecules at _____ $^\circ C$?

Options:

(a) $-100^\circ C$

(b) $-253^\circ C$

(c) $-20^\circ C$

(d) $-235^\circ C$

Answer: (b)

Solution:

$$V_{rms} = \sqrt{\frac{3RT}{M}}$$

$$\sqrt{\frac{3R \times 320}{32}} = \sqrt{\frac{3R \times T}{2}}$$

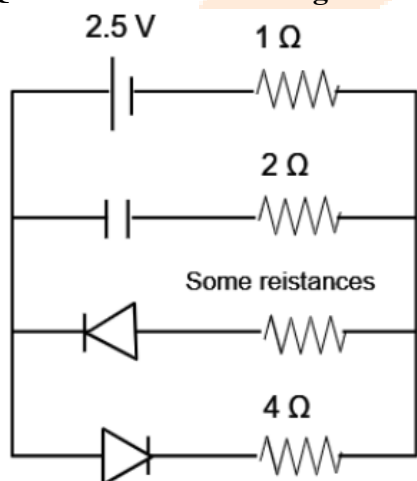
$$\Rightarrow \frac{310}{32} = \frac{T}{2}$$

$$\Rightarrow T = 20K$$

$$T_c = (20 - 273)^{\circ}C$$

$$= -253^{\circ}C$$

Question: Find the charge on capacitor in steady state?



Answer: $(10 \mu C)$

Solution:

- Since Diode in CD is reverse biased so, CD can be ignored.
- At steady state no current flows through Capacitor so AB also ignored.
- Now, from Fig $R_{eq} = 5\Omega$.

So, $I = \frac{2.5}{5} = 0.5 \text{ A}$.

To find charge through capacitor we require V_{AB} .

$\Rightarrow V_{AB} = V_{EF} = 4(0.5) = 2 \text{ V}$

$\therefore Q = 2 \text{ C}$

Question: Refractive index of right angled prism is $\sqrt{2}$. What should be the angle of incidence for a light ray such that emerging ray grazes out of the surface.

Answer: (90°)

Question: The ratio of Wavelength of 3rd line of Paschen Series, 2nd line of Balmer in H-atom

Options:

(a) $9/4$

(b) $4/9$

(c) $3/4$

(d) $4/3$

Answer: (a)

Solution:

$$\frac{1}{\lambda} = \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) Z^2 \quad (Z=1)$$

$$\frac{1}{\lambda_{\text{Paschen}}} = \left(\frac{1}{9} - \frac{1}{36} \right)$$

$$= \frac{36-9}{36 \times 9} = \frac{1}{12}$$

$$\boxed{\lambda_{\text{Paschen}} = 12}$$

$$\frac{1}{\lambda_{\text{Balmer}}} = \frac{1}{4} - \frac{1}{16}$$

$$= \frac{16-4}{16 \times 4}$$

$$\frac{1}{\lambda_{\text{Bal}}} = \frac{3}{16}$$

$$\boxed{\lambda_B = \frac{16}{3}}$$

$$\boxed{\frac{\lambda_P}{\lambda_B} = \frac{12}{16} \times 3 = \frac{9}{4}}$$

Question: If mass of the particle $m = 2 \text{ kg}$ and displacement of particle varies with the time as $x = \alpha t^2 + \beta t + \gamma$. Then find the work done in the time interval $t = 2 \text{ sec}$ to $t = 3 \text{ sec}$ ($\alpha = 1, \beta = 1, \gamma = 1$)

Answer: (24 J)

Solution:

$$V = \frac{dx}{dt} = 2\alpha t + \beta = 2t + 1$$

$$V_i = 5 \text{ m/s}, \quad V_f = 7 \text{ m/s}.$$

Work Energy theorem,

$$W = \Delta KE \quad \left(KE = \frac{1}{2} m v^2 \right)$$

$$= \frac{1}{2} m (V_f^2 - V_i^2)$$

$$= \frac{1}{2} \times 2 (7^2 - 5^2) \Rightarrow 24 \text{ J}$$

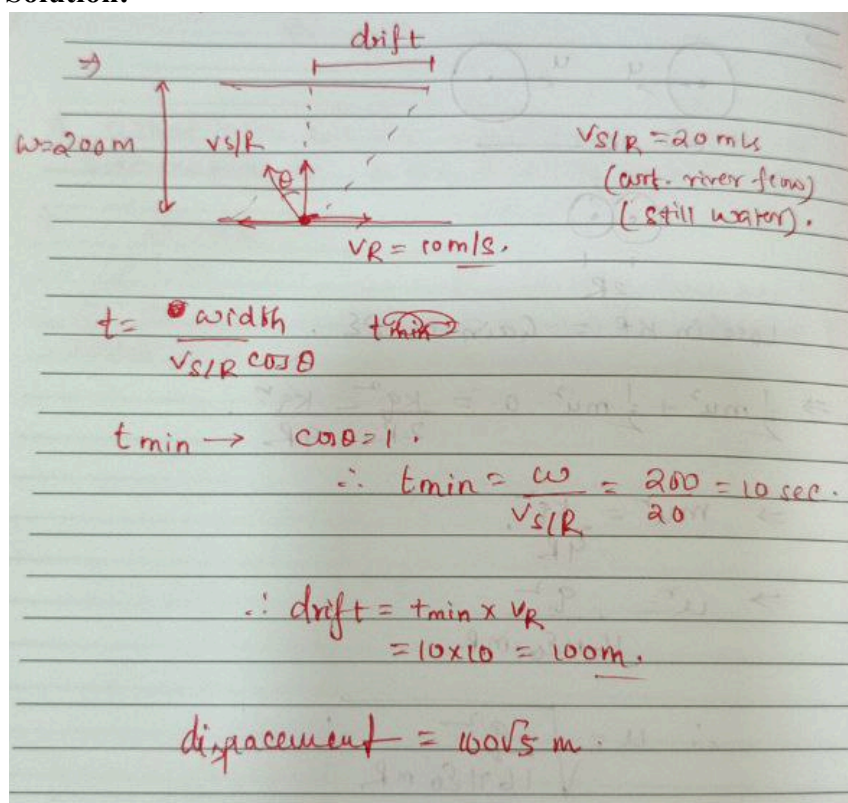
Question: Width of river is 200 m flowing with velocity 10 ms. A boat can move with speed of 20 m/sec. Wrt river flow. Find min time to cross river and displacement along river bank.

Options:

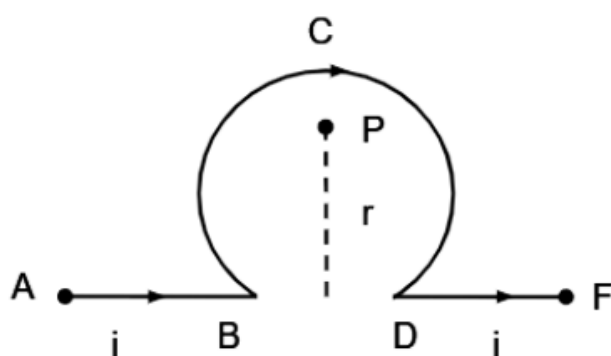
- (a) $100\sqrt{5}$ m
- (b) $10\sqrt{5}$ m
- (c) 100 m
- (d) 105 m

Answer: (a)

Solution:



Question: Current 'i' is passing through the given arrangement of conductors. Find 'B' at point 'P'.



Answer: $B_P = \frac{\mu_0 i}{2\pi r} [1 - \pi] \odot \text{Tesla}$

Solution:

$$B_P = B_{AB} + B_{BCD} + B_{DE}$$

$$= \frac{\mu_0 i}{2\pi r} - \frac{\mu_0 i}{2r} + \frac{\mu_0 i}{4\pi r}$$

$$= \frac{\mu_0 i}{2\pi r} - \frac{\mu_0 i}{2r}$$

$$\therefore B_P = \frac{\mu_0 i}{2\pi r} [1 - \pi] \odot \text{ Tesla}$$

(AB and DE are infinite wires and BCD is circular loop)

$$\begin{cases} B_{AB} \Rightarrow \odot \\ B_{BCD} \Rightarrow \otimes \\ B_{DE} \Rightarrow \odot \end{cases}$$

Question: In isobaric expansion work is 100 J. Find heat given to the gas ($\gamma = 1.4$).

Options:

- (a) 300
- (b) 350
- (c) 400
- (d) 450

Answer: (b)

Solution:

$$W = 100 \text{ J} = n R \Delta T$$

$$\Delta Q = ?$$

$$\Delta Q = n C_p \Delta T$$

$$= \left(\frac{f}{2} + 1 \right) n R \Delta T$$

$$= \left(\frac{f}{2} + 1 \right) \times 100$$

$$= \left(\frac{1}{2 \times 0.2} + 1 \right) \times 100$$

$$= \left(\frac{1}{0.4} + 1 \right) \times 100$$

$$= \left(1 + 0.4 \right) \times 100 = \frac{1.4 \times 100}{0.4}$$

$$= \frac{140}{0.4} = \frac{1400}{4} = 350$$

$\gamma = 1 + \frac{2}{f}$
 $1.4 = 1 + \frac{2}{f}$
 $\Rightarrow \frac{2}{f} = 0.4$
 $\Rightarrow f = \frac{2}{0.4}$
 $= \frac{2}{0.2}$

JEE-Main-21-01-2026 (Memory Based)
[EVENING SHIFT]
Chemistry

Question: 1 g of an organic compound produce 1.49 of $\text{Mg}_2\text{P}_2\text{O}_7$ Determine % of P

Options:

- (a) 42
- (b) 55
- (c) 24
- (d) 38

Answer: (a)

Question: Given below are two statements

Statement-I: The correct order for radius is $\text{Al} > \text{Mg} > \text{Mg}^{2+} > \text{Al}^{3+}$

Statement-II: Atomic size always, depends on electronegativity.

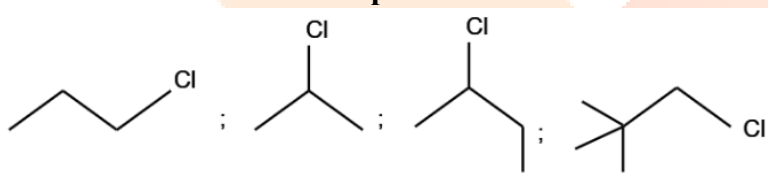
In the light of the above statements, choose the correct option.

Options:

- (a) Statement-I and II are correct
- (b) Both Statement-I and II are incorrect
- (c) Statement-I is correct but Statement-II incorrect
- (d) Statement-I is incorrect but Statement-II correct

Answer: (b)

Question: Following 4 molecules are given and among then, one is optically effective :
Find % carbon in that compound:-



Options:

- (a) 52
- (b) 65
- (c) 35
- (d) 89

Answer: (a)

Question: When 8.74 g MnO_2 is treated with HCl , then what will be the weight of $\text{Cl}_2(\text{g})$ obtained?

Molar mass of $\text{MnO}_2 = 87.4 \text{ g/mol}$

Options:

- (a) 7.1 g
- (b) 17.1 g

- (c) 14.2 g
(d) 3.55 g

Answer: (a)

Question: Some species are given

Ni^{2+} , Fe^{2+} , Co^{2+} , V^{3+} and Ti^{2+}

How many species has magnetic moment (spin only) less than 3 BM.

Options:

- (a) 4
(b) 2
(c) 3
(d) 1

Answer: (c)

Question: Find concentration of X^{2-} at equilibrium in 0.1 M H_2X .

Given $K_{a_1} = 2.5 \times 10^{-7}$

$K_{a_2} = 1 \times 10^{-13}$

Options:

- (a) 2.5×10^{-7}
(b) 1×10^{-13}
(c) 6×10^{-12}
(d) 5×10^{-10}

Answer: (b)

Question: Correct Nucleophilicity order,

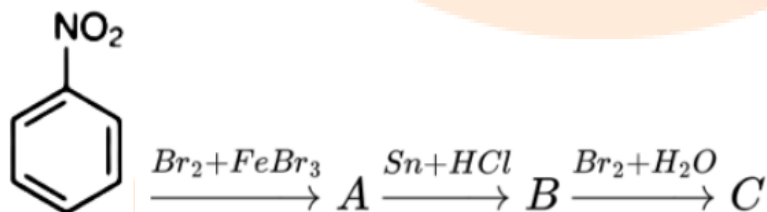
CH_3COO^- , $\text{Ph}-\text{O}^-$, OH^- , ClO_4^-

Options:

- (a) $\text{ClO}_4^- < \text{CH}_3\text{COO}^- < \text{PhO}^- < \text{OH}^-$
(b) $\text{CH}_3\text{COO}^- < \text{PhO}^- < \text{OH}^- < \text{ClO}_4^-$
(c) $\text{ClO}_4^- < \text{CH}_3\text{COO}^- < \text{OH}^- < \text{PhO}^-$
(d) $\text{ClO}_4^- < \text{PhO}^- < \text{OH}^- < \text{CH}_3\text{COO}^-$

Answer: (a)

Question: Number in bromine in final product



Options:

- (a) 2
(b) 4
(c) 1
(d) 3

Answer: (b)

Question: Match the following

Reagent	Reaction
A) H_2 , Pd-BaSO ₄	i) Etard Reaction
B) SnCl_2 , HCl	ii) Rosenmund Reduction
C) CrO_2Cl_2 , CS ₂	iii) Gatterman-Koch Reaction
D) CO, HCl Anhyd. AlCl ₃	iv) Stephan Reduction

Options:

- (a) A-(ii), B- (iv), C-(i), D-(iii)
- (b) A-(iii), B- (ii), C-(iv), D-(i)
- (c) A-(iv), B- (i), C-(ii), D-(iii)
- (d) A-(ii), B- (iii), C-(i), D-(iv)

Answer: (a)

Question: What will be the ratio of wavelength of 3rd line at Paschen Series to 2nd line of Balmer series of H-atom?

Options:

- (a) $\frac{4}{9}$
- (b) $\frac{3}{2}$
- (c) $\frac{2}{3}$
- (d) $\frac{16}{4}$

Answer: (a)

Question: $\text{K}_2\text{Cr}_2\text{O}_7$ is heated with KCl in pressure of H_2SO_4 . Find the correct match of product with their oxidation state.

Options:

- (a) CrO_2Cl_2 , +6
- (b) $\text{Cr}_2\text{O}_2\text{Cl}_2$, +6
- (c) $\text{Cr}_2\text{O}_2\text{Cl}_2$, +5
- (d) CrO_2Cl_2 , +5

Answer: (a)

Question: Osmotic pressure of a solution is 12 atm. What is the concentration of NaCl solution which is isotonic to the given solution at 900 K.

$R = 0.082 \text{ Lit-atm K}^{-1} \text{ mol}^{-1}$

Assume 100% dissociation.

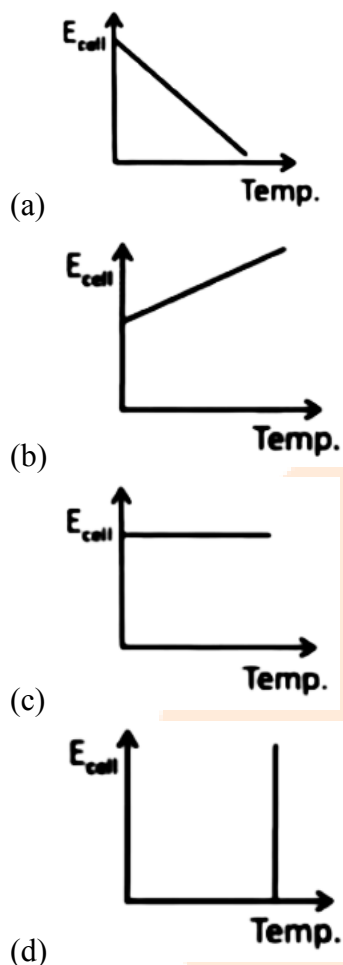
Options:

- (a) 0.4878 M
- (b) 0.02439 M
- (c) 0.2439 M
- (d) 0.04878 M

Answer: (c)

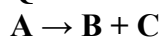
Question: Find out correct Graph

Options:



Answer: (a)

Question: For first order kinetics reaction,



If initial pressure of A is 1 bar and at time 100 s, the total pressure is 1.5 bar, then find the rate constant of the reaction.

Options:

- (a) $6.93 \times 10^{-3} \text{ s}^{-1}$
- (b) $6.93 \times 10^{-2} \text{ s}^{-1}$
- (c) 0.693 s^{-1}
- (d) 6.93 s^{-1}

Answer: (a)

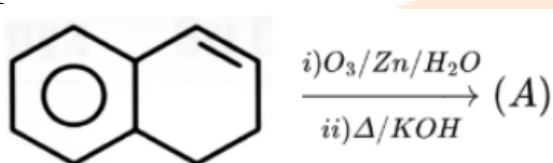
Question: Match the list-I with list-II

Reagent	Reaction
A) Cis 2-butene, Trans 2-butene	P) Functional Isomer
B) Butanoic acid, Isopropyl methanoate	Q) Stereoisomer

C) 1-butene, 2-butene	R) Positional Isomer
D) n-pentane, isopentane	S) Chain isomer

Options:

- (a) A-(P); B-(Q); C-(R); D-(S)
 (b) A-(P); B-(Q); C-(S); D-(R)
 (c) A-(Q); B-(P); C-(S); D-(R)
 (d) A-(Q); B-(P); C-(R); D-(S)

Answer: (d)**Question:****Find the product A****Options:**

- (a)
- (b)
- (c)
- (d)

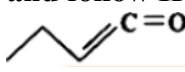
Answer: (a)**Question:** Given below are two statements.**Statement-I:** The correct order of electron gain enthalpies $\text{Cl} > \text{F} > \text{Br} > \text{I}$.**Statement-II:** $\text{SnCl}_6 > \text{SnCl}_4$, $\text{PbCl}_4 > \text{PbCl}_2$, $\text{UF}_6 > \text{UF}_4$, order of Ionic Character. which is the correct option.**Options:**

- (a) Both Statement-I and statement-II are correct
 (b) Both statement-I and statement-II are incorrect
 (c) Statement-I is correct and statement-II is incorrect
 (d) Statement-i is incorrect and statement-II is correct

Answer: (b)

Question: Given below are two statements.

Assertion: In aromatic Compound should be cyclic and planar with all C - should be sp^2 with one unhybrid p-orbitals and follow Huckels rules.

Reason: In given compound,  2 carbon is sp^3 & 1 is sp^2 .

Options:

- (a) Both (A) and (R) are true, and (R) is the correct explanation (A).
- (b) Both (A) and (R) are true, but (R) is NOT the correct explanation of (A).
- (c) (A) is true, but (R) is false.
- (d) (A) is false, but (R) is true.

Answer: (b)

Question: Solubility product of $MX(s)$ is 10^{-10} and $E_{\frac{M^+}{M}}^0 = 0.71 V$. Find out $E_{M/MX/X^-}^0$.

Options:

- (a) 0.119 V
- (b) -0.119 V
- (c) 1.301 V
- (d) -1.301 V

Answer: (b)

Question: Give the correct bond length order of C—H; C—O; C=O; CN

Options:

- (a) C - H > C-O > C = O > CN
- (b) C - O > C = O > CN > C-H
- (c) C-H > C-O > CN > C = O
- (d) CN > C - O > C - H > C = O

Answer: (a)

Question: Select correct statements

- (A) Amylose a long branched chain polymer, formed by $C_2 - C_6$ glycosidic linkage.
- (B) During denaturation destroyed but primary structures remained intact.
- (C) Globular proteins structures results when the chains of polypeptides coil around to give a spherical shape.
- (D) Carbohydrates which reduce Fehling's and Tollen's reagent are reducing sugars.

Options:

- (a) A, B, C, D
- (b) A, C, D
- (c) B, C, D
- (d) C, D

Answer: (c)

JEE-Main-21-01-2026 (Memory Based)
[EVENING SHIFT]
MATHS

Question: Let $f(x) = x^3 + x^2 f'(1) + 2xf''(2) + f'''(3)$, $x \in \mathbb{R}$. Then the value of $f'(5)$ is

Options:

- (a) $\frac{117}{5}$
 (b) $\frac{62}{5}$
 (c) $\frac{657}{5}$
 (d) $\frac{2}{5}$

Answer: (a)

$$f'(1) = A \quad f''(2) = B \quad f'''(3) = C$$

$$f(x) = x^3 + Ax^2 + 2xB + C$$

$$f'(x) = 3x^2 + 2Ax + 2B$$

$$f''(x) = 6x + 2A$$

$$f'''(x) = 6 \Rightarrow f'''(3) = 6 = C$$

$$C = 6$$

$$f''(2) = 12 + 2A$$

$$B = 12 + 2A \dots\dots(i)$$

$$f'(1) = A \Rightarrow A = 3 + 2A + 2B$$

$$\Rightarrow -A - 2B = 3 \dots\dots(ii)$$

From (i) & (ii)

$$A = -\frac{27}{5} \quad \& \quad B = \frac{6}{5}$$

$$f'(5) = 3(5)^2 + 2 \times -\frac{27}{5} \times 5 + 2 \times \frac{6}{5}$$

$$= 75 - 54 + \frac{12}{5} = 21 + \frac{12}{5}$$

$$= \frac{117}{5}$$

Question: The largest $n \in \mathbb{N}$, for which $7n$ divides $101!$

Options:

- (a) 16
 (b) 18
 (c) 19
 (d) 15

Answer: (a)

$$\left[\frac{101}{7} \right] + \left[\frac{101}{7^2} \right]$$

$$= 14 + 2$$

$$= 16$$

Question: If three vectors are given as shown.

If angle between vectors $\vec{p}$ and $\vec{q}$ is θ where $\cos \theta = \frac{1}{\sqrt{3}}$ and $|\vec{p}| = 2\sqrt{3}$, $|\vec{q}| = 2$.

$$|\vec{p} \times (\vec{q} - 3\vec{r})|^2 - 3|\vec{r}|^2$$

Then the value of

Answer: (92)

$$\vec{p} + \vec{r} = \vec{q}$$

$$\vec{r} = \vec{q} - \vec{p}$$

$$|\vec{p} \times (\vec{q} - 3\vec{q} + 3\vec{p})|^2 - 3|\vec{q} - \vec{p}|^2$$

$$|-2\vec{p} \times \vec{q}|^2 - 3|\vec{q} - \vec{p}|^2$$

$$4|\vec{p}|^2 |\vec{q}|^2 \sin^2 \theta - 3(|\vec{q}|^2 + |\vec{p}|^2 - 2|\vec{p}||\vec{q}| \cos \theta)$$

$$= 4 \times 12 \times 4 \left(1 - \frac{1}{3}\right) - 3\left(4 + 12 - 2 \times 2\sqrt{3} \times \frac{1}{\sqrt{3}}\right)$$

$$= 16 \times 12 \times \frac{2}{3} - 3(16 - 4)$$

$$= 16 \times 8 - 3 \times 12$$

$$= 128 - 36 = 92$$

Question: Let one end of a focal chord of the parabola $y^2 = 16x$ be $(16, 16)$. If $P(\alpha, \beta)$ divides this focal chord internally in the ratio $5 : 2$; then the minimum value of $\alpha + \beta$ is equal to:

Options:

- (a) 7
- (b) 22
- (c) 5
- (d) 16

Answer: (a)

$$4a = 16 \Rightarrow a = 4$$

$$2at_1 = 16 \Rightarrow t_1 = 2$$

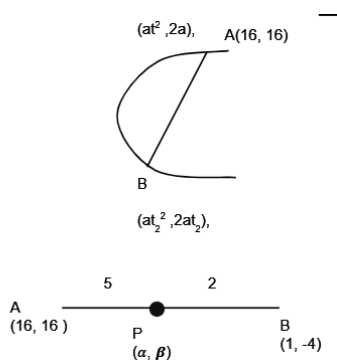
$$t_1 t_2 = -1 \Rightarrow t_2 = -\frac{1}{2}$$

$$B = (1, -4)$$

$$AP : PB = 5 : 2$$

$$\alpha = \frac{5+32}{7}, \beta = \frac{-20+32}{7}$$

$$\alpha + \beta = \frac{37+12}{7} = 7$$



Question: If area bounded by curves $y \leq 1-2x$ and $y \leq 4-x^2$, $x \geq 0$, $y \geq 0$ is equal to $\frac{\alpha}{\beta}$ where G.C.D. $(\alpha, \beta) = 1$ then $\alpha + \beta$ is

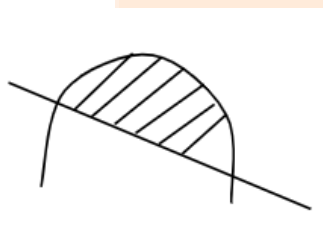
Answer: $(\frac{20\sqrt{5}}{3})$

Solving $y = -2x$, $y = 4 - x^2$

$$x^2 - 2x - 4 = 0$$

$$x = \frac{2 \pm \sqrt{4+16}}{2} = 1 + \sqrt{5}$$

$$\begin{aligned} \text{Area} &= \frac{d^3}{6l} = \frac{(2\sqrt{5})^3}{6 \times 1} = \frac{2 \times 5\sqrt{5}}{6} \\ &= \frac{20\sqrt{5}}{3} \end{aligned}$$



Question: If $\frac{dy}{dx} \sec x - 2y = 2 + 3 \sin x$ and $y(0) = -\frac{7}{4}$ then find $y\left(\frac{\pi}{6}\right)$

Answer: $(\frac{-5}{2})$

$$\frac{dy}{dx} - (2 \cos x)y = (2 + 3 \sin x) \cos x$$

$$I. F = e^{-2 \sin x}$$

$$ye^{-2 \sin x} = \int e^{-2 \sin x} (2 + 3 \sin x) \cos x \, dx$$

$$= \int e^{-2} (2 + 3t) \, dt, \, t = \sin x$$

$$= (2 + 3t) \cdot \frac{e^{-2t}}{-2} - \int 3 \cdot \frac{e^{-2t}}{-2} \, dt$$

$$= -\frac{1}{2} (2 + 3t) e^{-2t} - \frac{3}{4} e^{-2t} + c$$

$$ye^{-2 \sin x} = -\frac{1}{2} (2 + 3 \sin x) e^{-2 \sin x} - \frac{3}{4} e^{-2 \sin x} + c$$

$$\text{If } x = 0, y = \frac{-7}{4}, \frac{-7}{4} = -1 - \frac{3}{4} + c \Rightarrow c = 0$$

$$\text{If } x = \frac{\pi}{6}, ye^{-1} = -\frac{1}{2} (2 + \frac{3}{2}) e^{-1} - \frac{3}{4} e^{-1}$$

$$y = \frac{-7}{4} - \frac{3}{4} = \frac{-5}{2}$$

Question: If $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 29 & 49 \\ 1 & 2 \end{bmatrix}$ satisfy $(A^{15} + B) \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

then which of the following values of (x, y) satisfies above equation.

Options:

(a) $x = 11, y = 2$

(b) $x = 5, y = 7$

(c) $x = 18, y = 11$

(d) $x = 16, y = 3$

Answer:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 2 & -4 \\ 1 & -2 \end{pmatrix}$$

$$p^2 = \begin{pmatrix} 2 & -4 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 2 & -4 \\ 1 & -2 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\Rightarrow p^n = 0 \, \forall n \geq 2$$

$$A^{15} = (I + p)^{15}$$

$$= I + {}^{15}C_1 p + {}^{15}C_2 p^2 + \dots + {}^{15}C_{15} p^{15}$$

$$A^{15} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + 15 \begin{pmatrix} 2 & -4 \\ 1 & -2 \end{pmatrix}$$

$$A^{15} = \begin{pmatrix} 31 & -60 \\ 15 & -29 \end{pmatrix}$$

$$(A^{15} + B) \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 60 & -11 \\ 16 & -27 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$60x - 11y = 0$$

$$16x - 27y = 0$$

$$x = 0 \text{ \& } y = 0$$

Question: $a_1, \frac{a_2}{2}, \frac{a_3}{2^2}, \dots, \frac{a_{10}}{2^9}$ are in GP having common ratio $\frac{1}{\sqrt{2}}$ and $a_1 + a_2 + a_3 + \dots + a_{10} = 62$ then find a_1

Answer: $(2(\sqrt{2} - 1))$

$$\frac{\frac{a_2}{2}}{a_1} = \frac{1}{\sqrt{2}} \Rightarrow \frac{a_2}{a_1} = \sqrt{2}$$

$$\frac{a_1 \left[(\sqrt{2})^{10} - 1 \right]}{\sqrt{2} - 1} = 62 \Rightarrow a_1 = 2(\sqrt{2} - 1)$$

Question: Let $f(x) = \lim_{n \rightarrow \infty} \left(\frac{1}{n^3} \sum_{k=1}^n \left[\frac{k^2}{3^x} \right] \right)$, where $[.]$ denotes the greatest integer function, then $12 \sum_{j=1}^{\infty} f(j)$ is equal to

Options:

- (a) 2
- (b) 3
- (c) 4
- (d) 1

Answer: (a)

$$[A] = A - \{A\}$$

$$f(x) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n^3} \cdot \frac{k^2}{3^x} - \sum_{k=1}^n \frac{1}{n^3} \cdot \left\{ \frac{k^2}{3^x} \right\}$$

$$\because \frac{1}{n^3} \rightarrow 0$$

$$\text{and } \left\{ \frac{k^2}{3^x} \right\} \in (0, 1)$$

$$f(x) = \frac{1}{3^x} \lim_{n \rightarrow \infty} \frac{1^2 + 2^2 + \dots + x^2}{n^3}$$

$$= \frac{1}{3^x} \lim_{n \rightarrow \infty} \frac{n(n+1)(3n+1)}{6n^3}$$

$$= \frac{1}{3^x} \times \frac{2}{6} = \left(\frac{1}{3} \right)^{x+1}$$

$$12 \sum_{i=1}^{\infty} f(j) = 12 \sum_{i=1}^{\infty} \left(\frac{1}{3} \right)^{j+1}$$

$$= 12 \left[\left(\frac{1}{3} \right)^2 + \left(\frac{1}{3} \right)^3 + \dots + \infty \right]$$

$$= 12 \left[\frac{\frac{1}{9}}{1 - \frac{1}{3}} \right] = 12 \times \frac{1}{9} \times \frac{3}{2}$$

$$= 12 \times \frac{1}{3 \times 2} = 2$$

Question: The number of solutions of equation $\tan 3x = \cot x$ in $x \in [0, 2\pi]$ is

Options:

- (a) 4
- (b) 6
- (c) 2
- (d) 8

Answer: (d)

$$\tan 3x = \tan\left(\frac{\pi}{2} - x\right)$$

$$3x = n\pi + \frac{\pi}{2} - x$$

$$\Rightarrow x = (2n + 1)\frac{\pi}{8}, 0 \leq n \leq 7$$

$$= 8$$

